CSCI3230 / ESTR3108 2026–27 First Term Assignment 2

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University Guideline on Academic Honesty:
`http://www.cuhk.edu.hk/policy/academichonesty/`
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Note. All computations were performed in Python (NumPy / scikit-learn); the hand calculations of Questions 1, 3 and 4 were independently cross-checked against the numerical results.

Question 1: SVM optimization and prediction

Data. The training set contains $m = 8$ points $\{(\mathbf{x}_i, y_i)\}_{i=1}^{8}$ and the test set contains 8 points:

	Training set								Test set							
i	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8
x_1	3.5	5	3	5.5	4.5	3.5	3	4	5.5	3.5	4.5	2.5	3.5	2.5	4	5
x_2	−2	1.5	3	−1	−5	−6.5	−4	−3.5	0.5	−0.5	1.5	−1	−4.5	−4	−6	−5
y	1	1	1	1	−1	−1	−1	−1	1	1	1	1	−1	−1	−1	−1

(a) Dual problem, support vectors, optimal hyperplane

Primal problem. For a hard-margin SVM we solve

$$\min_{\mathbf{w},b} \ \frac{1}{2}\|\mathbf{w}\|^2 \qquad \text{s.t.} \qquad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1, \quad i = 1, \ldots, 8.$$

Dual problem. Introducing Lagrange multipliers $\alpha_i \geq 0$ for the constraints, the Lagrangian is

$$\mathcal{L}(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2}\|\mathbf{w}\|^2 - \sum_{i=1}^{8} \alpha_i \big[y_i(\mathbf{w}^\top \mathbf{x}_i + b) - 1\big].$$

Setting $\partial\mathcal{L}/\partial\mathbf{w} = 0$ and $\partial\mathcal{L}/\partial b = 0$ gives

$$\mathbf{w} = \sum_{i=1}^{8} \alpha_i y_i \mathbf{x}_i, \qquad \sum_{i=1}^{8} \alpha_i y_i = 0,$$

and substituting back yields the dual problem

$$\boxed{\max_{\boldsymbol{\alpha}} \ \sum_{i=1}^{8} \alpha_i - \frac{1}{2}\sum_{i=1}^{8}\sum_{j=1}^{8} \alpha_i \alpha_j y_i y_j \, \mathbf{x}_i^\top \mathbf{x}_j \qquad \text{s.t.} \qquad \sum_{i=1}^{8} \alpha_i y_i = 0, \quad \alpha_i \geq 0 \ \forall i.}$$

Support vectors. Given $\alpha_1 = \alpha_8 = 0.8$ and $\alpha_2 = \cdots = \alpha_7 = 0$, the KKT complementarity condition $\alpha_i\big(y_i(\mathbf{w}^\top \mathbf{x}_i + b) - 1\big) = 0$ states that only points with $\alpha_i > 0$ lie on the margin. Hence the support vectors are

$$\boxed{\mathbf{x}_1 = (3.5, -2)^\top \text{ with } y_1 = +1 \qquad \text{and} \qquad \mathbf{x}_8 = (4, -3.5)^\top \text{ with } y_8 = -1.}$$

Optimal hyperplane. First,

$$\mathbf{w}^* = \sum_{i=1}^{8} \alpha_i y_i \mathbf{x}_i = 0.8 \cdot 1 \cdot \begin{pmatrix} 3.5 \\ -2 \end{pmatrix} + 0.8 \cdot (-1) \cdot \begin{pmatrix} 4 \\ -3.5 \end{pmatrix} = \begin{pmatrix} 2.8 \\ -1.6 \end{pmatrix} - \begin{pmatrix} 3.2 \\ -2.8 \end{pmatrix} = \boxed{\begin{pmatrix} -0.4 \\ 1.2 \end{pmatrix}}.$$

For the bias, using each support vector $y_s(\mathbf{w}^\top \mathbf{x}_s + b) = 1$:

$$\begin{aligned} \mathbf{x}_1: \quad & b = y_1 - \mathbf{w}^{*\top} \mathbf{x}_1 = 1 - \big[(-0.4)(3.5) + (1.2)(-2)\big] = 1 - (-1.4 - 2.4) = 4.8, \\ \mathbf{x}_8: \quad & b = y_8 - \mathbf{w}^{*\top} \mathbf{x}_8 = -1 - \big[(-0.4)(4) + (1.2)(-3.5)\big] = -1 - (-1.6 - 4.2) = 4.8. \end{aligned}$$

Both give $b^* = 4.8$ (and averaging the two gives the same value). The optimal hyperplane (function) is therefore

$$\boxed{f(\mathbf{x}) = \mathbf{w}^{*\top} \mathbf{x} + b^* = -0.4\, x_1 + 1.2\, x_2 + 4.8 = 0}$$

i.e. $x_2 = \frac{1}{3} x_1 - 4$, with margin $\frac{2}{\|\mathbf{w}^*\|} = \frac{2}{\sqrt{1.6}} \approx 1.581$.

(b) Predicting the test set

Given the optimal hyperplane, the prediction rule is the sign of the decision function:

$$\hat{y} = \operatorname{sign}\big(f(\mathbf{x})\big) = \operatorname{sign}\big(\mathbf{w}^{*\top} \mathbf{x} + b^*\big) = \operatorname{sign}\big(-0.4\, x_1 + 1.2\, x_2 + 4.8\big),$$

where $\hat{y} = +1$ means class $+1$ and $\hat{y} = -1$ means class -1. Evaluating:

Test point	(x_1, x_2)	$-0.4x_1 + 1.2x_2 + 4.8$	$\hat{y}$	y	correct?
1	$(5.5, 0.5)$	$-2.2 + 0.6 + 4.8 = 3.2$	$+1$	$+1$	yes
2	$(3.5, -0.5)$	$-1.4 - 0.6 + 4.8 = 2.8$	$+1$	$+1$	yes
3	$(4.5, 1.5)$	$-1.8 + 1.8 + 4.8 = 4.8$	$+1$	$+1$	yes
4	$(2.5, -1)$	$-1.0 - 1.2 + 4.8 = 2.6$	$+1$	$+1$	yes
5	$(3.5, -4.5)$	$-1.4 - 5.4 + 4.8 = -2.0$	-1	-1	yes
6	$(2.5, -4)$	$-1.0 - 4.8 + 4.8 = -1.0$	-1	-1	yes
7	$(4, -6)$	$-1.6 - 7.2 + 4.8 = -4.0$	-1	-1	yes
8	$(5, -5)$	$-2.0 - 6.0 + 4.8 = -3.2$	-1	-1	yes

So the predictions are $\boxed{+1,\ +1,\ +1,\ +1,\ -1,\ -1,\ -1,\ -1}$, i.e. test points 1–4 are classified as class $+1$ and points 5–8 as class -1, giving 100% test accuracy.

(c) Removing training points

- **Remove the 2nd training point** $(5, 1.5, +1)$. This point has $\alpha_2 = 0$, i.e. it is not a support vector and does not lie on the margin. Removing a non-support vector does not change the optimal solution: the remaining dual problem (with the same $\alpha_1 = \alpha_8 = 0.8$) still yields $\mathbf{w}^* = (-0.4, 1.2)^\top$, $b^* = 4.8$, hence **the test predictions do not change**.

- **Remove the 8th training point** $(4, -3.5, -1)$. This point is a support vector ($\alpha_8 = 0.8 > 0$), so the margin hyperplane must change. Re-solving the SVM on the remaining 7 points (numerically, exact solution) gives a different classifier $\mathbf{w}^* \approx (0.2353, 0.9412)^\top$, $b^* \approx 2.0588$. *However*, the decision boundary still separates the two classes in exactly the same way for the test points: all eight test points are still classified correctly, i.e. **the test predictions also do not change** (100% accuracy is retained). This is because the margin is supported by the point $\mathbf{x}_1 = (3.5, -2, +1)$ and the new closest negative point $\mathbf{x}_5 = (4.5, -5, -1)$, which yields a classifier whose decision region covers all test points as before.

- *Numerical verification:* removing point 2 gives exactly $\mathbf{w}^* = (-0.4, 1.2)^\top$, $b^* = 4.8$; removing point 8 gives $\mathbf{w}^* = (0.2353, 0.9412)^\top$, $b^* = 2.0588$; in both cases the predictions on the test set are identical to those in part (b).

Question 2: SVM with kernels on the attached dataset

Data loading

The data were loaded exactly with the suggested snippet:

```
import pandas as pd
train = pd.read_csv('path/to/training.csv', encoding='utf-8-sig')
test = pd.read_csv('path/to/test.csv', encoding='utf-8-sig')
X_train, y_train = train.iloc[:, :-1].to_numpy().astype(float), train.iloc[:, -1].
    to_numpy().astype(float)
X_test, y_test = test.iloc[:, :-1].to_numpy().astype(float), test.iloc[:, -1].to_numpy().
    astype(float)
```

Models

Model 1: linear kernel (soft-margin, $C = 1.0$). This is the baseline given in the assignment. The linear kernel $\kappa(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^\top \mathbf{x}_j$ finds a straight-line decision boundary; with $C = 1.0$ it tolerates some margin violations. It achieves only $\approx 51.02\%$ test accuracy — the data are not linearly separable.

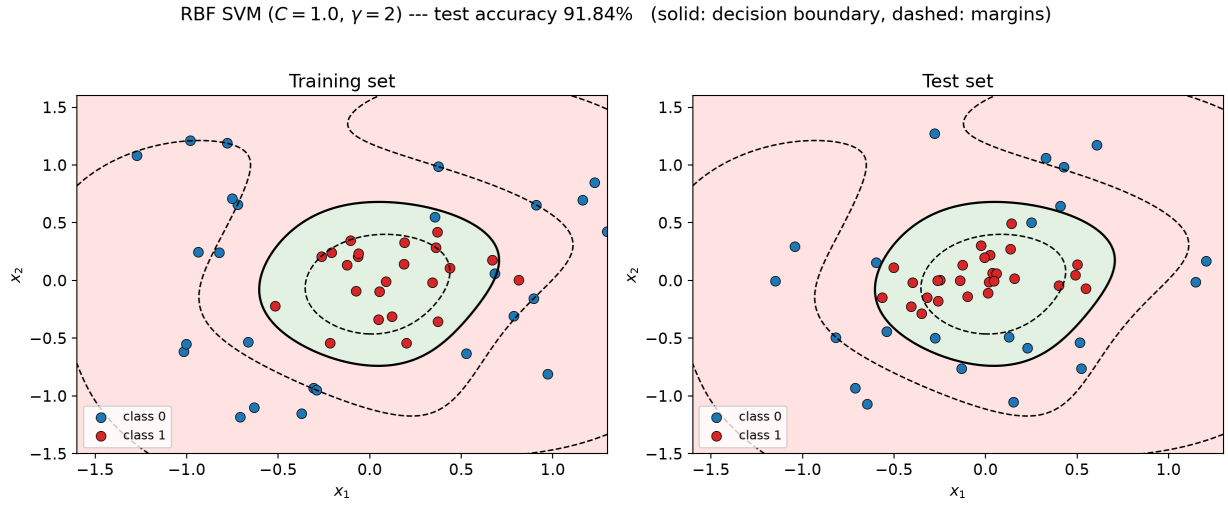
Model 2: RBF (Gaussian) kernel (soft-margin, $C = 1.0$, $\gamma = 2$). The radial basis function kernel $\kappa(\mathbf{x}_i, \mathbf{x}_j) = \exp\big(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2\big)$ maps the data into an infinite-dimensional feature space, so a non-linear (roughly circular) decision boundary can be found. With $C = 1.0$ and $\gamma = 2$ the model has 23 support vectors and reaches **91.84% test accuracy**, well above the required 85%.

I also tried polynomial ($\approx 67.3\%$) and sigmoid ($\approx 46.9\%$) kernels, and a grid of C values (0.1, 0.5, 1, 2, 5, 10, 50) and of γ (0.1–10) for RBF; the best accuracy 91.84% was reached by several RBF settings, e.g. $C = 1$, $\gamma = 2$ (23 SVs) and $C = 2$, $\gamma = 1$ (19 SVs), so $C = 1.0$, $\gamma = 2$ is reported.

Plots

Decision boundary (solid line) and margins (dashed lines) on both the training and test sets:

Code



RBF SVM ($C = 1.0$, $\gamma = 2$): decision boundary (solid) and margins (dashed) shown on the training set (left) and test set (right). Test accuracy = 91.84%.

```
from sklearn.svm import SVC

# Model 1: linear kernel, soft margin
clf_lin = SVC(kernel='linear', C=1.0)
clf_lin.fit(X_train, y_train)
acc_lin = clf_lin.score(X_test, y_test) # 0.5102

# Model 2: RBF kernel, soft margin
clf_rbf = SVC(kernel='rbf', C=1.0, gamma=2)
clf_rbf.fit(X_train, y_train)
acc_rbf = clf_rbf.score(X_test, y_test) # 0.9184
print(len(clf_rbf.support_)) # 23 support vectors
```

The full plotting script (decision boundary + margins on both sets) is included as `q2_svm.py`.

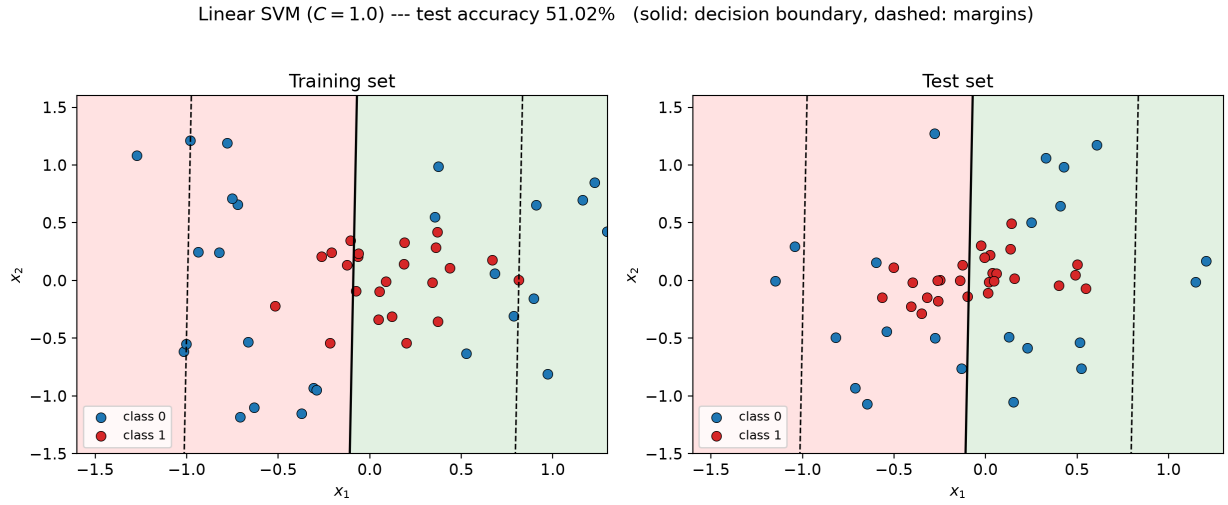
Question 3: K-means and DBSCAN

Points: $a(1,2)$, $b(2,3)$, $c(2,4)$, $d(6,5)$, $e(3,1)$, $f(3,3)$, $g(4,4)$, $h(7,6)$, $i(8,8)$, $j(8,9)$, $k(9,4)$, $l(5,9)$, $m(4,7)$, $n(6,8)$, $o(6,4)$.

(a) K-means, first iteration

Initial centres: $\mu_1 = (7,4)$, $\mu_2 = (3,6)$, $\mu_3 = (7,9)$. Using squared Euclidean distance, assign each point to the nearest centre:

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
d_1^2 (to $(7,4)$)	40	26	25	2	25	17	9	4	17	26	4	29	18	17	1
d_2^2 (to $(3,6)$)	20	10	5	10	25	9	5	16	29	34	40	13	2	13	13
d_3^2 (to $(7,9)$)	85	61	50	17	80	52	34	9	2	1	29	4	13	2	26
cluster	2	2	2	1	1	2	2	1	3	3	1	3	2	3	1



Linear SVM ($C = 1.0$): decision boundary (solid) and margins (dashed). Test accuracy = 51.02%, matching the example in the assignment.

(Ties: e is equally distant from centres 1 and 2 — assigned to cluster 1 here; the tie can be broken arbitrarily without changing the following analysis.)

Updated centres after the first iteration:

$$\mu_1^{(1)} = \text{mean}\{d, e, h, k, o\} = \left(\tfrac{6+3+7+9+6}{5},\ \tfrac{5+1+6+4+4}{5}\right) = \boxed{(6.2,\ 4)},$$

$$\mu_2^{(1)} = \text{mean}\{a, b, c, f, g, m\} = \left(\tfrac{1+2+2+3+4+4}{6},\ \tfrac{2+3+4+3+4+7}{6}\right) = \boxed{(2.6667,\ 3.8333)},$$

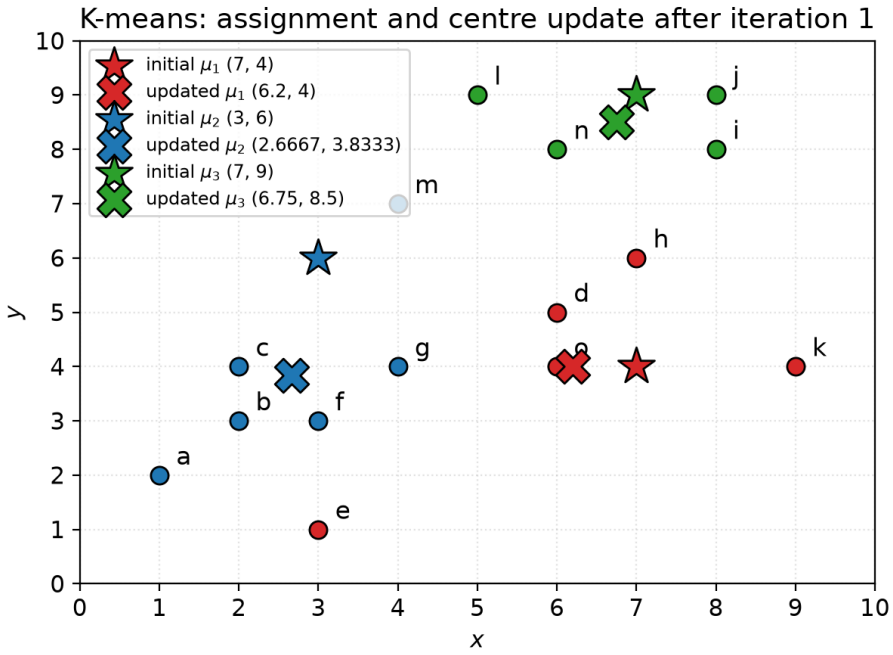
$$\mu_3^{(1)} = \text{mean}\{i, j, l, n\} = \left(\tfrac{8+8+5+6}{4},\ \tfrac{8+9+9+8}{4}\right) = \boxed{(6.75,\ 8.5)}.$$

(b) DBSCAN: $\epsilon = 2$, minpts $= 3$

ϵ-neighbourhoods (including the point itself, Euclidean distance):

point	N_ϵ	point	N_ϵ
a	$\{a, b\}$	i	$\{i, j, n\}$
b	$\{a, b, c, f\}$	j	$\{i, j\}$
c	$\{b, c, f, g\}$	k	$\{k\}$
d	$\{d, h, o\}$	l	$\{l, n\}$
e	$\{e, f\}$	m	$\{m\}$
f	$\{b, c, e, f, g\}$	n	$\{i, l, n\}$
g	$\{c, f, g, o\}$	o	$\{d, g, o\}$
h	$\{d, h\}$		

- **Core points** ($|N_\epsilon| \geq 3$): $\boxed{b,\ c,\ d,\ f,\ g,\ i,\ n,\ o}$
- **Border points** (not core, but within ϵ of a core point): $\boxed{a,\ e,\ h,\ j,\ l}$
- **Noise points** (neither core nor border): $\boxed{k,\ m}$



Initial centres (stars) and updated centres after iteration 1 (X marks).

(c) Is h density-reachable from b?

Yes. Density-reachability requires a chain of points $p_0, \ldots, p_k$ with $p_0 = b$, $p_k = h$, where each step goes from a core point to a point in its ϵ-neighbourhood. One such chain is

$$\boxed{b \to c \to g \to o \to d \to h}$$

Check: $c \in N_\epsilon(b)$, $g \in N_\epsilon(c)$, $o \in N_\epsilon(g)$, $d \in N_\epsilon(o)$, $h \in N_\epsilon(d)$, and b, c, g, o, d are all core points. Hence h is density-reachable from b.

(d) Density-based clusters

The core points form two connected components (via direct density-reachability between core points):

$$\{b, c, d, f, g, o\} \cup \text{border } \{a, e, h\} \qquad \text{and} \qquad \{i, n\} \cup \text{border } \{j, l\}.$$

Therefore

- Cluster 1: $\boxed{\{a,\ b,\ c,\ d,\ e,\ f,\ g,\ h,\ o\}}$
- Cluster 2: $\boxed{\{i,\ j,\ l,\ n\}}$

and $k,\ m$ remain noise points (not part of any cluster).

Question 4: Hierarchical clustering (group average)

Distance matrix:

	A	B	C	D	E
A	0	8	6	2	1
B	8	0	7	3	6
C	6	7	0	9	5
D	2	3	9	0	4
E	1	6	5	4	0

Group average distance between clusters U, V: $d(U,V) = \frac{1}{|U||V|} \sum_{u \in U} \sum_{v \in V} d(u,v)$.

Iteration 1. Smallest distance is $d(A,E) = 1$. Merge A and E into AE ($d = 1$). Updated table:

	AE	B	C	D
AE	0	(8+6)/2 = 7	(6+5)/2 = 5.5	(2+4)/2 = 3
B	7	0	7	3
C	5.5	7	0	9
D	3	3	9	0

Iteration 2. Smallest distances: $d(B,D) = 3$ and $d(AE,D) = 3$ (tie; tie-breaking does not affect the final structure). Merge B and D into BD ($d = 3$). Updated table:

$$d(AE,BD) = \frac{8+6+2+4}{4} = \frac{20}{4} = 5, \qquad d(C,BD) = \frac{7+9}{2} = 8, \qquad d(C,AE) = 5.5.$$

	AE	BD	C
AE	0	5	5.5
BD	5	0	8
C	5.5	8	0

Iteration 3. Smallest distance is $d(AE,BD) = 5$. Merge into $AEBD$ ($d = 5$). Updated table:

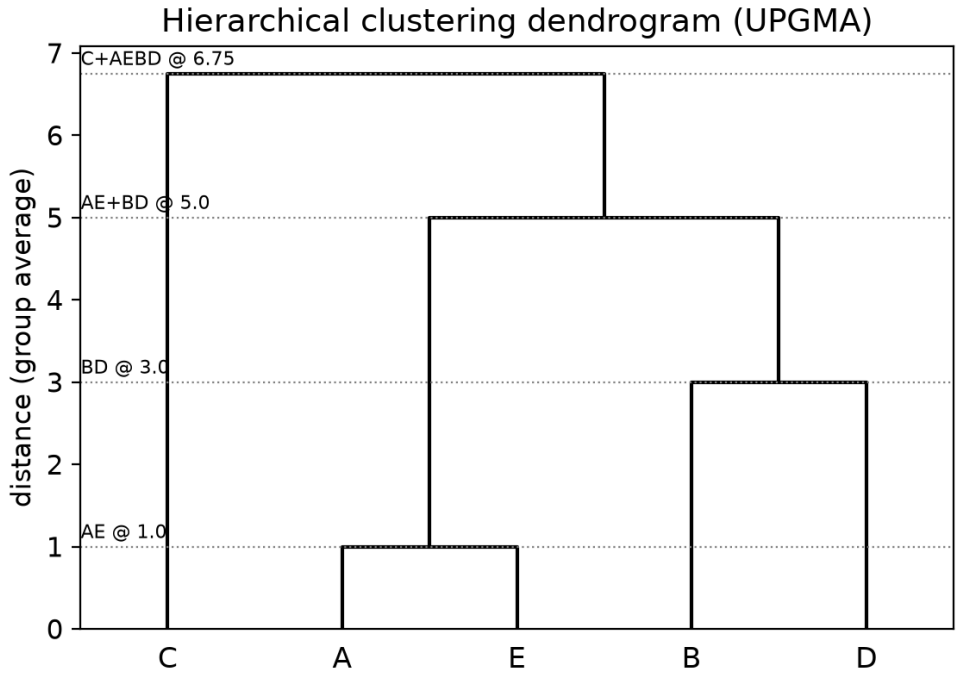
$$d(C,AEBD) = \frac{6+5+7+9}{4} = \frac{27}{4} = 6.75.$$

	AEBD	C
AEBD	0	6.75
C	6.75	0

Iteration 4. Merge $AEBD$ and C into the single cluster $ABCDE$ at $d = 6.75$. Done.

Summary of the merges (order in which points are merged):

$$\boxed{(A,E) \xrightarrow{1.0} (B,D) \xrightarrow{3.0} (AE,BD) \xrightarrow{5.0} (AEBD,C) \xrightarrow{6.75}}$$



Final dendrogram. Merge order: A–E at 1, B–D at 3, AE–BD at 5, then C joins at 6.75.